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the probability of the visitor not seeing a heron on any given day is 1 − 0.2 = 0.8. Since seeing a swan is independent of seeing a heron, not seeing a swan is likewise independent of not seeing a heron. Thus, the probability that a visitor to the pond will see neither a swan nor a heron on any given day is 0.65 × 0.8 = 0.52.

The correct answer is *0.52*.

•

Two primary school teachers are debating the merits of classrooms with highly decorated walls.

Teacher 1: A recent study has shown that highly decorated walls can interfere with learning in the classroom. We should therefore decrease the number of wall decorations in our classrooms.

Teacher 2: The results of that study are correct, but pertain only to the retention of information from single lessons in which the students were in the classroom for only a short period of time. Our students are in the same classroom all day, five days a week. Without some added stimulation, students can become bored and therefore even more distracted than they would be by highly decorated walls. I therefore maintain that highly decorated walls serve to increase learning over the long term.

Select for X and for Y the options such that the following statement, if filled in accordingly, has the most support from the information provided. Make only two selections, one in each column. Teacher 2 suggests that he agrees with Teacher 1 about the effects of wall decorations on ____X____, but disagrees about the effects of wall decorations on ____Y____.

X	Y	
		retention of certain specified types of information
		the effectiveness of teaching methods that eschew rote learning
		the ability of students to maintain skills if practiced in various classrooms
		retention of information from multiple lessons given in the same room over a period of time
		retention of information from lessons given in rooms occupied for only a short time

Answer Explanation:

Infer

***X*:**

Teacher 1 cites a study as having shown that highly decorated walls interfere with classroom learning. Teacher 2 remarks that these study results are correct with regard to retention of information from single, brief lessons. Thus, the two teachers agree that wall decorations interfere with retention of information from lessons given in rooms occupied for only a short time.

The correct answer is *retention of information from lessons given in rooms occupied for only a short time.*

***Y*:**

By citing the study without qualification as a reason to remove classroom wall decorations, Teacher 1 implicitly accepts that the study results hold in general. Thus, Teacher 1 seems to believe that highly decorated walls generally interfere with classroom learning. But Teacher 2 argues that when students are in the same room for many lessons over long periods, decorated walls reduce their boredom and distraction, helping them learn better in the long run. So, the two teachers evidently disagree about whether highly decorated walls interfere with students' retention of information from multiple lessons given in the same room over time.

The correct answer is *retention of information from multiple lessons given in the same room over a period of time.*

•

Natalya put 14 blue marbles, 14 red marbles, and no other marbles into 3 empty cups. She put 2 blue marbles and 6 red marbles in Cup A, 6 blue marbles and 4 red marbles in Cup B, and 6 blue marbles and 4 red marbles in Cup C. After this, Dmitry will randomly pick three marbles, one from each cup.

Consistent with the given information, select for *3 blue* the probability that Dmitry will pick 3 blue marbles, and select for *3 red* the probability that Dmitry will pick 3 red marbles. Make only two selections, one in each column.

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3 blue	3 red	
		0.090
		0.120
		0.125
		0.150
		0.250
		0.500

Answer Explanation:

Apply

***3 blue*:**

Using the multiplication rule for independent events, the probability of selecting 3 blue marbles is the probability of selecting 1 blue marble from Cup A times the probability of selecting 1 blue marble from Cup B times the probability of selecting 1 blue marble from Cup C, which is equal to $\frac{\text{number of blue marbles in Cup A}}{\text{number of marbles in Cup A}}$

$\times \frac{\text{number of blue marbles in Cup B}}{\text{number of marbles in Cup B}}$

$\times \frac{\text{number of blue marbles in Cup C}}{\text{number of marbles in Cup C}} = \frac{\cancel{2}^{1}}{\cancel{8}^{4}} \times \frac{\cancel{6}^{3}}{\cancel{10}^{5}} \times \frac{\cancel{6}^{3}}{\cancel{10}^{5}} \times \frac{9}{100} = 0.09$

The correct answer is *0.090*.

***3 red*:**

Using the multiplication rule for independent events, the probability of selecting 3 red marbles is the probability of selecting 1 red marble from Cup A times the probability of selecting 1 red marble from Cup B times the probability of selecting 1 red marble from Cup C, which is equal to $\frac{\text{number of redmarbles in Cup A}}{\text{number of marbles in Cup A}}$

$\times \frac{\text{number of redmarbles in Cup B}}{\text{number of marbles in Cup B}} \times \frac{\text{number of redmarbles in Cup C}}{\text{number of marbles in Cup C}}$

$= \frac{\cancel{6}^{3}}{\cancel{8}^{1}} \times \frac{\cancel{4}^{1}}{10} \times \frac{4}{10} \times \frac{12}{100} = 0.12$

The correct answer is *0.120*.

•

In recent years, commercial beekeepers have reported *colony collapse:* the disappearance of entire colonies of bees. A biologist, Dr. B, argues that colony collapse is caused by a new parasite to which bees have not acquired immunity. A chemist, Dr. C, argues that colony collapse is caused by genetic damage from pesticides. And an ecologist, Dr. E, argues that 3 specific commercial beekeeping practices—transportation of hives over long distances, indiscriminate medication, and inbreeding—compromise both the immunity and the genetic diversity of bees, leaving them susceptible to mass die-offs from normally benign microbes.

Complete the table based on the stated views of Dr. B, Dr. C, and Dr. E. In the first column, select the factor common to both Dr. B's and Dr. E's accounts of colony collapse. In the second column, select the factor common to both Dr. C's and Dr. E's accounts of colony collapse. Make two selections, one in each column.

Dr. B and Dr. E	Dr. C and Dr. E	Factor
		Genetics
		Immunity
		Medication

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		New parasites
		Normally benign microbes
		Pesticides

Answer Explanation:

Strategize

By listing the available factors associated with the information given about the arguments by Dr. B, Dr. C, and Dr. E, we can determine which factor is common to Dr. B and Dr. E and which factor is common to Dr. C and Dr. E.

Dr. B: New parasites, Immunity

Dr. C: Genetics, Pesticides

Dr. E: Medication, Immunity, Genetics, Normally benign microbes

***Dr. B and Dr. E*:**

The factor common to both Dr. B and Dr. E is Immunity.

The correct answer is *Immunity*.

***Dr. C and Dr. E*:**

The factor common to both Dr. C and Dr. E is Genetics.

The correct answer is *Genetics*.

•

Martine is scheduling five contractors (Contractors A through E) to do various tasks during Monday through Friday of the upcoming week. For each day, exactly one contractor will be scheduled to work, and each contractor's task will be completed on the day for which that contractor is scheduled. The scheduling must conform to the following constraints:

Contractor A must be scheduled to work on Monday.

Contractor B must be scheduled to work exactly two days after Contractor A.

Contractor E must be scheduled to work exactly three days after Contractor C.

Statement: Given the constraints, Martine must schedule Contractor B to work on ___1___ and Contractor D to work on ___2___.

Select for *1* and for *2* the options that complete the statement so that it is accurate based on the information provided. Make only two selections, one in each column.

1	2	
		Monday
		Tuesday
		Wednesday
		Thursday
		Friday

Answer Explanation:

Recognize

Using the first two constraints, Contractor A must be scheduled on Monday, and Contractor B must be scheduled two days later, which is on Wednesday.

Monday	Tuesday	Wednesday	Thursday	Friday
A		B		

According to the third constraint, Contractor C can only be scheduled on Monday or Tuesday, since three days later needs to be within the same Monday-through-Friday week. Since Contractor A must be scheduled on Monday, it follows that Contractor C must be scheduled on Tuesday, hence Contractor E must be scheduled three days later,

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follows that Contractor C must be scheduled on Tuesday; hence Contractor E must be scheduled three days later, which is on Friday.

Monday	Tuesday	Wednesday	Thursday	Friday
A	C	B		E

Finally, the remaining contractor, Contractor D, must be scheduled on the remaining day, Thursday, since each day exactly one contractor will be scheduled.

Monday	Tuesday	Wednesday	Thursday	Friday
A	C	B	D	E

***1*:**

Using the last table above, Contractor B must be scheduled on Wednesday.

The correct answer is *Wednesday*.

***2*:**

Using the last table above, Contractor D must be scheduled on Thursday.

The correct answer is *Thursday*.

Questions 428 to 450 — Difficulty: Medium

428. Supervisor's memo: At the present rate of manufacture we will not have the 1,200 circuit boards assembled in time to ship them to the customer. We currently have 10 workers, all working 8 hours a day at the same hourly rate, assembling the boards, and to date they've assembled 400. In order to finish the customer's order in time, I'll need to have each of the workers work an additional 2 hours for each of the 10 workdays between now and the deadline.

One can determine the hourly rate at which each worker assembled circuit boards up to the date of the supervisor's memo by dividing ___1___ circuit boards by the product of ___2___ worker-hours per day and the number of days since the workers began assembling the circuit boards.

Select *Circuit boards* for the option that fills the blank labeled 1 in the given statement, and select *Worker-hours* for the option that fills the blank labeled 2 in the given statement to create the most accurate statement on the basis of the information provided. Make only two selections, one in each column.

Circuit boards	Worker-hours	
		80
		400
		500
		800
		1,000
		1,200

Answer Explanation:

Strategize

***Circuit boards*:**

The number of circuit boards assembled is the product of the number of workers, the hourly rate per worker, the number of hours worked per day, and the number of days worked. Using the given information, it follows that

(400 boards) = (10 workers)(hourly rate) (8 hours/day)(number of days)

Now solve for the hourly rate:

$$\begin{aligned}\text{hourly rate} &= \frac{400\ boards}{(10\ workers)(8\ hours)(number\ of\ days)} \\ &= \frac{400\ boards}{(80\ worker-hours)(number\ of\ days)}\end{aligned}$$

The correct answer is *400*.

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***Worker-hours*:**

The analysis in Response 1 shows that the number of worker-hours per day is 80.

The correct answer is *80*.

429. In an experiment, one thousand nine-year-old children were allowed to choose whether to participate in a program in which the researchers taught them dance during the daily break from their lessons. Four hundred of the children chose to participate for at least one year. At the end of the year, researchers found that the children who had participated had significantly better balance, on average, than those who had not. The researchers hypothesized that dancing resulted in a sustained improvement in the children's sense of balance.

It would be most helpful in evaluating the researchers' hypothesis to know whether the researchers ___1___ prior to having ___2___.

Select *Researchers* for the phrase that fills the blank labeled 1 in the given statement, and select *Prior to* for the phrase that fills the blank labeled 2 in the given statement to create the most accurate statement on the basis of the information provided. Make only two selections, one in each column.

Researchers	Prior to	
		tested the children's ability to dance
		designed a second experiment
		divided the children into the two experimental groups
		tested the children's sense of balance
		taught dance to the children through the dance program

Answer Explanation:

Evaluate

***Researchers* and *Prior to*:**

Note that because of the relationship between the two answers that complete the sentence, the responses have to be considered together, not separately.

The question asks what would be most helpful to know when evaluating the researchers' hypothesis. To answer this, it is useful to consider some alternative explanations for the better balance of the children who had participated in the program. One alternative would be that the children who had decided to participate in the program already had better balance before participating. Therefore, it would be useful to know whether they actually did have any improvement in their balance. To resolve that issue, it would be helpful if we knew that the researchers had this information, which they could have acquired by testing the children's sense of balance prior to teaching them dance through the dance program.

The correct answer to Response 1 is *tested the children's sense of balance*.

The correct answer to Response 2 is *taught dance to the children through the dance program*.

430. Klaus: Closing our neighborhood's school will mean a major hassle for neighborhood families. Students will have to travel farther to and from school, which means they'll have less time for homework. That will likely affect them academically. Additionally, a vacant school building in our neighborhood is sure to attract crime and drag down home values.

Rena: Our neighborhood's school never had a great reputation anyway. Closing it means students will get to go to a larger school—with a better reputation—just a few kilometers away, where they'll have more opportunities to participate in extracurricular activities. That should make homes here more valuable to potential buyers. And the vacant school building could be converted into a much-needed community center.

Select *Klaus* for the phrase that best describes a factor that is specifically addressed by Klaus but not by Rena, and select *Rena* for the phrase that best describes a factor that is specifically addressed by Rena but not by Klaus. Make only two selections, one in each column.

Klaus	Rena	
		The vacant school building
		The distance students must travel to school

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		Extracurricular activities
		Real estate values
		Students' academic performance

Answer Explanation:

Recognition

***Klaus*:**

Klaus and Rena both specifically mention the vacant school building, the distance students must travel to school, and real estate values. Klaus, unlike Rena, also specifically mentions students' academic performance. (It is possible that when Rena talks about the schools' reputations, the schools' students' academic performance is one of the factors that affects the schools' reputations, but we do not know for certain.)

The correct answer is *Students' academic performance*.

***Rena*:**

Klaus and Rena both specifically mention the vacant school building, the distance students must travel to school, and real estate values. Rena, unlike Klaus, also specifically mentions extracurricular activities.

The correct answer is *Extracurricular activities*.

431. Alfredo: The United States could make significant progress toward becoming independent of petroleum-derived gasoline as a source of energy by diverting a large proportion of corn grown in the US to the production of ethanol for fuel. Brazil is virtually independent of petroleum as a source of fuel because that country has been very successful in ethanol production.

Mavis: Ethanol derived from corn will never be a viable substitute for petroleum-derived gasoline in the US. While it is true that Brazil has been very successful in replacing gasoline with ethanol, Brazil's ethanol is derived from sugarcane, which is very easy to convert to ethanol and, in Brazil, very easy to cultivate. Corn, on the other hand, is difficult to grow, requiring intensive cultivation. Also, the amount of power that can be garnered from the land is much less per square meter for corn ethanol.

In the table, select the statement about which Alfredo and Mavis would most clearly agree and select the statement about which Alfredo and Mavis would most clearly disagree. Make only two selections, one in each column.

Agree with each other about	**Disagree with each other about**	
		Corn ethanol produces much less power per square meter of cultivated land than does sugarcane ethanol.
		Ethanol is easier to produce from sugarcane than from corn.
		There are no crops, other than corn, for producing ethanol that could viably be grown in the US.
		Brazil has been very successful at replacing petroleum with ethanol.
		Producing corn ethanol is a viable strategy for a nation intent on reducing dependence on petroleum.

Answer Explanation:

Infer

***Agree with each other about*:**

Alfredo and Mavis both accept the claim that Brazil has been very successful at replacing petroleum with ethanol.

The correct answer is *Brazil has been very successful at replacing petroleum with ethanol*.

***Disagree with each other about*:**

Whereas Alfredo believes that Brazil's success at replacing demand for petroleum with ethanol supports the

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claim that producing corn ethanol is a viable strategy for a nation intent on reducing demand on petroleum, Mavis does not. Mavis points out that producing ethanol from sugarcane, as Brazil does, has very significant advantages over producing ethanol from corn.

The correct answer is *Producing corn ethanol is a viable strategy for a nation intent on reducing dependence on petroleum*.

432. Radhika is refinancing a business loan and is considering 2 different loan offers. Under Offer 1, the loan's initial principal would be $190,000, and she would pay down $1,250 in principal with each monthly payment during the first year of the loan. Under Offer 2, $4,000 in refinancing fees would be added to bring the principal to $194,000, but she would pay down $1,775 in principal with each monthly payment during the first year of the loan.

In the first column of the table, select the amount of principal that would remain after 12 monthly payments under Offer 1. In the second column of the table, select the amount of principal that would remain after 12 monthly payments under Offer 2. Make only two selections, one in each column.

Principal remaining after 12 months under Offer 1	**Principal remaining after 12 months under Offer 2**	
		$168,700
		$171,000
		$172,700
		$175,000
		$176,700
		$179,000

Answer Explanation:

Apply

***Principal remaining after 12 months under Offer 1*:**

Under Offer 1, the principal would be $190,000, and the monthly payment would be $1,250. The total of 12 monthly payments would be 12($1,250) = $15,000. Therefore, the principal remaining after 12 months under Offer 1 would be $190,000 − $15,000 = $175,000.

The correct answer is *$175,000*.

***Principal remaining after 12 months under Offer 2*:**

Under Offer 2, with the $4,000 refinancing fee added, the principal would be $194,000 and the monthly payment would be $1,775. The total of 12 monthly payments would be 12($1,775) = $21,300. Therefore, the principal remaining after 12 months under Offer 2 would be $194,000 − $21,300 = $172,700.

The correct answer is *$172,700*.

433. Professor A: The field of Artificial Intelligence (AI) was originally aimed exclusively at the construction of *thinking machines*, that is, computer systems with human-like general intelligence. But the difficulty of this goal of general AI has led researchers to shift their focus to producing narrow AI systems: software displaying intelligence regarding specific tasks in relatively narrow domains such as chess. The problem is, narrow AI and general AI are fundamentally and entirely different pursuits. If modeling general intelligence is the objective, AI researchers must redirect themselves toward the original goals of the field and confront the more difficult issues of human-like intelligence.

Professor B: Narrow AI work is producing real results today, such as chess-playing programs that can defeat any human, programs that translate speech to text, and programs that can drive automated vehicles. Achievements in this field will continue to flow, and eventually narrow AI will lead to the creation of systems with general intelligence.

On the basis of the given information, select a claim that *Professor A* would most likely *disagree* with, and select a claim that *Professor B* would most likely *disagree* with. Select only two claims, one in each column.

Professor A disagrees	**Professor B disagrees**	

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		The creation of computer systems that display human-like general intelligence is an important goal.
		The creation of systems that display intelligence regarding specific tasks is an especially important step toward the creation of systems with general intelligence.
		Systems with general intelligence are more difficult to create than are systems with narrow intelligence.
		Research in the narrow AI field is not likely to lead to significant breakthroughs in the general AI field.
		The original aim of AI research was to create systems with general intelligence.
		The research into narrow AI is not producing results that real people can benefit from today.

Answer Explanation:

Infer

***Professor A disagrees*:**

Professor A states that the field of AI (Artificial Intelligence) originally was intended to create thinking machines, that is, computers that could display human-like general intelligence. Professor A then states that over time the field of AI shifted its focus to narrow AI systems, ones that address relatively narrow realms of intelligence, such as chess playing. Professor A states that narrow AI will not aid in being able to produce systems that model general intelligence, because narrow AI and general AI are entirely different pursuits. Therefore, a claim that Professor A would *disagree* with is that the creation of systems that display intelligence regarding specific tasks is an especially important step toward the creation of systems with general intelligence.

The correct answer is *The creation of systems that display intelligence regarding specific tasks is an especially important step toward the creation of systems with general intelligence*.

***Professor B disagrees*:**

Unlike Professor A, Professor B believes that the achievements of narrow AI—of which Professor B says there are many—will eventually lead to the creation of systems with general intelligence. Therefore, a claim that Professor B would *disagree* with is that research in the narrow AI field will not likely lead to significant breakthroughs in the general AI field.

The correct answer is *Research in the narrow AI field is not likely to lead to significant breakthroughs in the general AI field.*

434. Swamp sparrows live in a variety of wetland habitats. Unlike most swamp sparrows, which live in freshwater habitats, the coastal-plain subspecies lives in tidal wetlands, where freshwater and seawater mix and the mud is gray rather than brown. Coastal-plain swamp sparrows differ from all other populations of swamp sparrows in having plumage that is gray-brown rather than rusty brown. DNA analysis indicates several important genetic differences between swamp sparrows that inhabit tidal marshes and other subspecies of swamp sparrows. Therefore, there must have been genetic-selection pressure on swamp sparrows in tidal marshes to become darker and grayer.

Select *Strengthen* for the statement that would, if true, most strengthen the argument, and select *Weaken* for the statement that would, if true, most weaken the argument. Make only two selections, one in each column.

Strengthen	Weaken	
		None of the genetic differences that have been identified in the genomes of coastal-plain swamp sparrows and freshwater swamp sparrows affect plumage color.
		Mud in tidal marshes tends to be grayish because of the presence of iron sulfide, whereas freshwater mud is browner because of the presence of iron oxide.
		Some species of birds that live in tidal marshes do not have gray plumage.
		The diets of both coastal-plain and freshwater swamp sparrows can change significantly from season to season.

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		Baby birds of the coastal-plain subspecies and baby birds of a freshwater swamp subspecies, all raised on an identical diet under controlled conditions, grew plumage similar in color to that of their respective parents.

Answer Explanation:

Evaluate

The passage discusses differences among various types of swamp sparrows and their habitats. Most swamp sparrows, which have rusty brown plumage, live in freshwater habitats, where the mud tends to be brown, but coastal-plain swamp sparrows, which have gray-brown plumage, live in tidal wetlands, where the mud tends to be gray. Research on the DNA of these types of birds has discovered important genetic differences between the coastal-plain swamp sparrows and other, freshwater-dwelling swamp sparrows. The passage draws the conclusion from these facts that there was some genetic pressure on coastal-plain swamp sparrows to become darker and grayer.

***Strengthen*:**

The argument concludes that the difference in plumage between tidal-marsh swamp sparrows and other swamp sparrows is genetic rather than, for instance, the direct environment in which an individual sparrow grows up or the sorts of food an individual sparrow consumes. Therefore, the argument in the passage would be strengthened if there were evidence that baby birds of the coastal-plain subspecies and baby birds of the freshwater species had plumage similar in color to their respective parents, even if the baby birds were raised in identical environments and fed the same sorts of food.

The correct answer is *Baby birds of the coastal-plain subspecies and baby birds of a freshwater swamp subspecies, all raised on an identical diet under controlled conditions, grew plumage similar in color to that of their respective parents*.

***Weaken*:**

Part of the evidence for the conclusion that the differences in color are due to genetic pressure is that DNA evidence shows important genetic differences among these subspecies. The argument would therefore be weakened if none of those genetic differences that have been identified has an effect on the color of plumage.

The correct answer is *None of the genetic differences that have been identified in the genomes of coastal-plain swamp sparrows and freshwater swamp sparrows affect plumage color*.

435. Boppo is a game played by 2 teams for a fixed duration (number of minutes). It is played with 8 members of each team on the floor—actually playing the game—at all times. At any time during the game, exactly one player from each team must be off the floor—not actually playing. Thus, by taking a break, by entering the game after it starts, or by leaving before it finishes, every player on each team must spend exactly 5 minutes resting or otherwise not playing.

In the table, identify a number of players per team and a number of minutes per game that are consistent with the given information about Boppo. Make only two selections, one in each column.

Number of players	Number of minutes	
		9
		11
		18
		30
		45
		50

Answer Explanation:

Apply

***Number of players*:**

Having 8 + 1 = 9 players per team is consistent with the requirement that, at any time during the game, 8 players must be on the floor actually playing the game and 1 player must be off the floor—not actually playing the game.

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game.

The correct answer is *9*.

***Number of minutes*:**

A fixed duration of 9 × 5 = 45 minutes per game is consistent with each of the 9 members on a team being required to spend exactly 5 minutes during each game resting or otherwise not playing the game.

The correct answer is *45*.

436. Loan X has a principal of $\$x$ and a yearly simple interest rate of 4%. Loan Y has a principal of $\$y$ and a yearly simple interest rate of 8%. Loans X and Y will be consolidated to form Loan Z with a principal of $\$(x+y)$ and a yearly simple interest rate of $r\%$, where $r = \dfrac{4x+8y}{x+y}$. Select a value for x and a value for y corresponding to a yearly simple interest rate of 5% for the consolidated loan. Make only two selections, one in each column.

x	*y*	*Value*
		21,000
		32,000
		51,000
		64,000
		81,000
		96,000

Answer Explanation:

Apply

The task is to determine values of x and y such that $\dfrac{4x+8y}{x+y} = 5$.

As shown, algebraic manipulation results in a simple relationship between x and y.

$\dfrac{4x+8y}{x+y} = 5$	given
$4x+8y = 5(x+y)$	multiply both sides by $x+y$
$4x+8y = 5x+5y$	apply the distributive property
$4x+3y = 5x$	subtract $5y$ from both sides
$3y = x$	subtract $4x$ from both sides

***x*:**

Choose a value for x that is 3 times another value in the *Value* column. Since the value for x will be greater than the other value, it is wise to start with the greatest value, which is 96,000. Since 96,000 is 3(32,000), 96,000 is a value for x that is consistent with the given information.

The correct answer is *96,000*.

***y*:**

From the explanation above, the value of y is 32,000.

The correct answer is *32,000*.

437. Perry, Maria, and Lorna are painting rooms in a college dormitory. Working alone, Perry can paint a standard room in 3 hours, Maria can paint a standard room in 2 hours, and Lorna can paint a standard room in 2 hours 30 minutes. Perry, Maria, and Lorna have decided that, to speed up the work, 2 of them will paint a standard room together.

Select the value closest to the shortest time in which a 2-person team could paint a standard room, and select the value closest to the longest time in which a 2-person team could paint a standard room, with each person working at his or her respective rate. Make only two selections, one in each column.

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Shortest time	Longest time	
		49 minutes
		1 hour 7 minutes
		1 hour 12 minutes
		1 hour 14 minutes
		1 hour 22 minutes
		1 hour 45 minutes

Answer Explanation:

Apply

***Shortest time*:**

Working together, Lorna and Maria can paint a standard room in the shortest amount of time because individually they can paint a standard room in less time than Perry. Lorna can paint a standard room in $2\frac{1}{2} = \frac{5}{2}$ hours, so she can paint $\frac{2}{5}$ room in 1 hour. Maria can paint a standard room in 2 hours, so she can paint $\frac{1}{2}$ room in 1 hour. Working together, they can paint $\frac{2}{5} + \frac{1}{2} = \frac{9}{10}$ room in 1 hour. It follows that they can paint 1 room in $\frac{10}{9}$ hours, which is equivalent to $1\frac{1}{9}$ hours or about 1 hour 7 minutes.

The correct answer is *1 hour 7 minutes*.

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